Agentic AI · Week 4 — follow-along sheet

Friday Oct 9, 2:30–4:30 · everything here lives at jiehou-lab.github.io/nrt_agent_week2

Today: one folder that reads papers, writes about them, and cites them correctly. Week 3 was about where a rule lives. Today the same three layers do real work on your own literature.

BEFORE YOU ARRIVE · please, at home

This is the only session with homework. It is ten minutes and doing it in the room costs everyone the interesting half.

A. Download and check

Course site → **Demo 10 · Paper writing with Zotero**
Unzip it, then:

```
cd Demo10_research_paper_with_zotero
tools/doctor.sh
```

Windows: install Git for Windows first — https://git-scm.com/download/win. It gives you git *and* **Git Bash**, and every tool in this package is a shell script. Run everything today from **Git Bash**, not PowerShell or CMD. macOS and Linux already have what they need.X

It prints what is missing and **the exact command to fix it**. Only the **Core** block must pass today: python3 and python-docx. Everything below Core is for the last hour.

```
pip3 install python-docx # if Core says it is missing
```

B. Zotero 7 + Better BibTeX

Install Zotero 7. Get the BBT .xpi from its GitHub releases page — in Firefox **right-click → Save Link As**, because a left-click installs it as a Firefox extension. Then **Zotero → Tools → Plugins → gear → Install Plugin From File**.

Careful: in Zotero 6 that menu was *Add-ons*. Zotero 7 renamed it to **Plugins**, and most tutorials still say the old name. This trips up nearly everyone.

Check it, with Zotero open:

```
curl -s http://127.0.0.1:23119/connector/ping
curl -s
'http://127.0.0.1:23119/better-bibtex/cayw?
probe=1'
```

The second must print ready.

C. Three papers, by DOI

In Zotero, the green **Add Item by Identifier** wand. Paste all three at once:

```
10.1038/s41598-023-49831-6
10.3390/jpm12050717
10.31083/RCM33425
```

All three are real and **CC BY 4.0**, so Zotero fetches the PDFs too. You may cite them.

D. MSU Overleaf — overleaf.com/edu/msu → **Log in with SSO**. MSU pays for premium, which includes Git. You will use it at 3:45.

F. LaTeX — optional, but do it the night before if you want it

Skip this if you want to. Overleaf compiles in the browser, and tools/tex-solo.sh falls back to text-only checks when there is no LaTeX — it still catches an unbalanced \begin{itemize}, a stray preamble, and a \cite key missing from your .bib, which are the three things that break somebody else's build. What it cannot then do is prove the page typesets.

Install it only if you want to see your own PDF locally. **Nobody is blocked without it.**

macOS — 92 MB, not MacTeX's 5 GB:

```
brew install --cask basictex
then open a NEW terminal, so PATH picks up
/Library/TeX/texbin
sudo tlmgr update --self
sudo tlmgr install latexmk natbib geometry
```

Windows — install **TeX Live** from https://tug.org/texlive/windows.html. It is large and slow to install, so start it before dinner. It bundles its own Perl, which matters: **MiKTeX does not, and latexmk is a Perl script** — MiKTeX users get The Perl interpreter could not be found and must install Strawberry Perl separately. LaTeX Workshop's own documentation recommends TeX Live for exactly this reason.X

Both — check it, in a new terminal:

```
latexmk -version
```

Using VS Code? Quit it completely after installing — Cmd-Q on Mac, not just closing the window. It inherits the PATH it was launched with, so a new terminal is not enough. This one step is why most people think the install failed.

```
tools/doctor.sh tells the three states apart: no
LaTeX, LaTeX present but not on PATH, and LaTeX
present but latexmk missing. Each prints its own
fix.
```

E. One line before an agent touches anything

```
git init && git add -A && git commit -m "before
the agent"
```

Still the most useful line in any of these demos.

PAPER NOTES · 2:30

1. Put a paper where the agent looks

```
cd Demo10_research_paper_with_zotero
claude
/agents
```

You should see **librarian, citecheck, drafter, prose**. Then drag one PDF from Zotero into papers/inbox/.

2. Turn it into a note

```
Use the librarian subagent on
papers/inbox/<file>.pdf
```

Watch the order it works in. Duplicate check on **DOI first** — not title, which gets retyped. Then papers/library/MANIFEST.md to find out where PDFs live. Then the note. Then one row in papers/_index.md. Then it rebuilds the view.

3. The checker that fixes nothing

```
python3 tools/check_notes.py
```

Ten required sections, a DOI, a matching index row. **It reports and repairs nothing — deliberately.** A checker that edits is one that can make its own findings disappear.

Your note's errors: _____ warnings: _____

4. Read the note like a reviewer

Open it. Two things:

- **## My take is empty.** It always is. No agent writes there. In six months it is the only section you can trust to be your own thinking.
- **## Problem and aim comes before ## Claim.** A note that opens with the claim has already accepted the paper's framing. What did they think was missing, and was the gap real?

5. Two more papers, then look at the shape

```
python3 tools/build_views.py
```

papers/notes/ is **flat** — no subfolders. A note carries area: in its frontmatter and that field holds **two** values; a folder holds one. So the generated view puts a two-area paper under **both** headings, which is the thing a folder cannot do.

The test before you ever add a subfolder: *can a paper be in two of these at the same time?* If yes, it is a field, not a folder.

```
Group papers/_index.md by area, 2024 onward,
flagged notes only.
```

No tool needed for a one-off slice. Just ask.

AGENTS AND SKILLS • 2:55

6. The same three layers as Week 3, doing real work

- **context** — CLAUDE.md — a **request**. Can be misread or forgotten.
- **permission** — .claude/settings.json — a **refusal**. The call is blocked.
- **capability** — a subagent's tools: list — an **absence**. Nothing to call.

Put every rule in the **lowest** layer that can hold it.

7. Prove the middle layer

```
cat .claude/settings.json
```

Find Edit(./papers/library/**) in the **deny** list. Then ask for it:

```
Edit papers/library/MANIFEST.md and add a
line saying "test".
```

Refused. "Do not edit the library" in CLAUDE.md is a request; this is a refusal. An agent quietly editing a reference library is how a decade of citations gets corrupted.

8. Prove the bottom layer

```
head -6 .claude/agents/citecheck.md
```

Look at the tools: line. **There is no Write.** Now:

```
Use the citecheck subagent on
papers/notes/<note>.md and fix anything wrong.
```

It reports. It cannot fix. **It did not decide to behave — there is no tool to ask about.**

9. Agent or skill?

- An **agent** is *who is working* — a role, with its own tool list and its own context window. librarian, citecheck, drafter, prose.
- A **skill** is *how this kind of work is done* — a procedure the main session follows. cite, lit-review, question, docx-review, manuscript-revision.

```
ls .claude/agents .claude/skills
```

Rule of thumb: needs a *narrower tool list* → agent. Needs a *consistent procedure* → skill.

10. Ask the library something it cannot answer

```
Use the question skill: do these models work
for patients who never reach a therapeutic INR?
```

The honest answer is **no source here separates that out**. You should get [GAP:] markers, not a synthesis.

If three papers produce a smooth, complete answer, something invented the rest. Noticing that is worth more than anything else today.

Gaps you got: _____

ZOTERO • 3:15

11. Export the library — the whole library

Right-click **My Library** → *Export Library* → **CSV**. Save as papers/library/library.csv.

Not a collection. A collection exports only its own items, while ~/Zotero/storage/ holds attachments for *everything*. You get a handful of CSV rows against hundreds of folders on disk, and every lookup outside the collection fails **with no error at all**.

```
head -1 papers/library/library.csv # must start:
Key,
wc -l papers/library/library.csv
```

12. The one vendor-specific file

```
cat papers/library/MANIFEST.md
```

Which tool, which metadata file, which key column, and how a key becomes a PDF path. **Every Zotero-specific fact in the workspace lives here.** Moving to Paperpile is this one file plus a re-export — no folder renamed, no agent edited.

13. Let Claude read your PDFs

Your PDFs are in ~/Zotero/storage/, outside the project. Add to .claude/settings.json:

```
"additionalDirectories":
["/Users/YOURNAME/Zotero/storage"]
```

Absolute path, **no tilde**. Restart claude. Now:

```
Use the citecheck subagent on
writing/drafts/<file>.md
```

It opens the actual PDFs. ***Cannot verify*** is a legitimate verdict and you will see several.

14. Two identifiers, two jobs

```
doi: "10.1038/s41598-023-49831-6" cross-tool.
survives migration.
item_key: "8KQW2MJ4" local to this one library.
```

Not the title, which gets retyped. Not a Better BibTeX cite key — **BBT regenerates it when you fix metadata**, silently breaking every citation already written.

Pin the ones you cite: the Zotero item → **Extra** → Citation Key: horne2012warfarin

15. The staleness trap

Add a paper to Zotero, forget to re-export, and library.csv quietly describes a library you no longer have.

```
python3 tools/check_notes.py
```

It compares the CSV against ~/Zotero/storage/ **both directions** and warns. For the LaTeX half there is a better answer, which is the next section.

OVERLEAF • 3:35

16. MSU pays for the part that matters

Every **paid** Overleaf plan includes Git; only the free plan does not. MSU's licence covers you. Check: open any project → **Menu → Sync**. *Git* listed → you have it.

17. Token, and where to keep it

Menu → Sync → Git → Generate token. Copy it immediately — it is shown once.

```
git config --global credential.helper osxkeychain
```

The token is the password. Your MSU SSO password will not work at the git prompt.
Do not paste the token into the URL.
https://git:TOKEN@git.overleaf.com/… works — and writes your token in plaintext into .git/config, where anything with repo access can read it, agents included. The keychain has no file to leak.

18. Clone, and mind which repo you are in

```
cd ~/.../research/writing/tex
git clone https://git@git.overleaf.com/<project-
id> mypaper
cd mypaper
cp ../../../papers/library/library.bib .
```

The mistake everyone makes once. Run git push one folder too high and you get fatal: No configured push destination. — writing/tex/ belongs to **your research repo**, which has no remote; the clone is the folder **inside** it. Two repos nested, and git silently acts on whichever encloses you. When unsure: **git remote -v**, and an Overleaf URL means you are in the right place.
19. The .bib, kept in sync automatically

Zotero → right-click **My Library** → *Export Library* → format **Better BibTeX** → **tick Keep updated**.

That tick is the whole difference. BBT re-exports whenever the library changes, so the .bib cannot drift. The CSV route has no equivalent.

Do not also use Overleaf's own Zotero button. Two .bib files with different keys for the same papers is a bug that reads like a typo for a very long time. Pick one generator.

20. Write a paragraph that rests on something

```
Use the drafter subagent: the Introduction's
second
paragraph, on whether published dose-model
accuracy
describes patients who never stabilise. Write
into
main.tex. LaTeX, \cite keys from library.bib.
```

Name the section, not the topic. "Write an introduction" produces the generic paragraph a model can write without reading anything.
Every claim comes back carrying its **tier**, and in LaTeX the non-citations sit in % comments so they never typeset:

```
... raised explained variance from 61.2% to 69.1%
.
% [GAP: error stratified by achieved INR band]
```

Leave the gaps in. They are the paper telling you what is still owed.
21. Compile — and the check that is the whole point

```
tools/tex.sh writing/tex/mypaper/main.tex --look
```

Measured on a real project: a **one-character typo** in a cite key (2012 → 2O12) gave

```
plain latexmk exit = 0 main.pdf built: YES
```

a PDF that looked finished, a **?** where the citation should be, and that reference **missing from the bibliography**. Nothing failed.
Overleaf's compiler will not fail on it either.
tex.sh makes it fatal, and names the key.

22. Look at the page

--look writes one PNG per page.

```
Read main-page-1.png. Did the citation render,
is the bibliography there, and does anything
run off the page?
```

This is the step people skip. A table running off a page limit, a figure separated from its caption, and a bibliography in the wrong style all compile perfectly.
23. Review, then push

```
Use the citecheck subagent on main.tex
Use the prose subagent on main.tex
```

Neither has a Write tool. They report; you decide. Then edit directly — this is LaTeX, no tracked-changes dance.

```
Show me git diff and summarise what changed,
one line per paragraph.
git add -A && git commit -m "..." && git push
```

Do not commit main.pdf — Overleaf compiles it, so a committed PDF is a stale binary that conflicts on every pull. The shipped .gitignore covers it.
Reload Overleaf. Your paragraph is there.

24. Next session, before anything

```
git pull
```

Not etiquette. A co-author editing in the browser makes Overleaf reject your push, and untangling that mid-session is the only part of this workflow that genuinely wastes time.

IF TIME · 4:15 — THE WORD ROUTE

Only if you write in Word. **LaTeX people are done** — go to *Your project*.
25. Markdown first, Word last

```
tools/render.sh rtf writing/drafts/<file>.md #
needs nothing
tools/render.sh live writing/drafts/<file>.md #
live Zotero fields
```

live needs a pandoc built **with HTTP** — Homebrew's and conda's are not, because pandoc 3.9 added a flag to build it without networking. Any pandoc 2.16.2–3.8 works. doctor.sh prints the fix.
26. In Word — three steps, and step 2 is the one nobody mentions

```
Zotero tab → Document Preferences → OK
Cursor under References → Add/Edit Bibliography
Refresh
```

Skip step 2 and Refresh runs, reports nothing, your citations show grey Zotero shading — **and no reference list appears**. The filter writes citation fields only; a bibliography is a different field only the Word plugin can insert.
27. Prove it is live, then send it back

Insert a citation — does the bibliography grow? Delete one — does it shrink? Change the style — do **all** citations restyle?

Then: **there is no round trip.** docx → md → docx returns **zero** field codes and loses **every image**. Measured. The file still opens and still looks right, which is what makes it dangerous.

```
python3 tools/docx_text.py <file>.docx # read it
python3 tools/docx_suggest.py <file>.docx
sug.json # tracked changes
```

The original is never written to. And for anything that came back from a co-author:

```
python3 tools/docx_text.py <new>.docx --against
<sent>.docx
```

Citation counts can match while a field is quietly damaged inside a deletion.

YOUR PROJECT · 4:25

Point this at your own literature. Nothing depends on the three demo papers.

LEVEL 1 — make it yours

- Rewrite the four areas in CLAUDE.md for your field. Four that still make sense in three years.
- Paste two paragraphs of your own writing into writing/STYLE.md — prose stops flagging your habits as faults.
- Write one real question in QUESTIONS.md. **State what evidence would answer it** — without that it is a topic, and a topic can never be closed.

LEVEL 2 — run it for a week

Ten papers in, ten notes out, one draft paragraph with its gaps visible. Then ask whether [GAP:] points at the same missing evidence three times — that is a research question forming.

LEVEL 3 — a real manuscript

Clone a real Overleaf project. Keep the .bib generated, never hand-edited. Decide whether Claude may git push on a co-authored paper — the package allows push, denies --force, and you may want that last step to stay yours.

The one sentence to remember. A note exists because you read the paper. Everything else — the keys, the index, the views, the checkers — is machinery for making that one fact useful three years later.